Department of Computer Science & Engineering
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Architectural Design Specification
CSE 4316: Senior Design I
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AccessVr
VR Wheelchair Controller

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1 Introduction

Access VR is a product that the aims to streamline and aid physical therapy. It allows users to be able to practice various scenarios and movements in a safe and controlled virtual space, thereby improving their confidence and skills in real-world wheelchair navigation. This will hopefully expedite the recovery and learning process of using a wheelchair for newly wheelchair bound users. Additionally this technology can be used to allow for wheelchair bounds users to have more accessibility in virtual environments as it replicates wheelchair movement more accurately than traditionally controllers.

1.1 Purpose and Use

The VR wheelchair system aims to provide an innovative training platform for wheelchair users to practice and improve their navigation skills within a virtual reality environment. By simulating real-world scenarios and obstacles, the system allows users to experience and adapt to various challenges without physical risk. This tool is intended to be used in rehabilitation centers, training facilities, and by individuals at home who seek to enhance their wheelchair maneuvering abilities. The system integrates with a physical wheelchair to provide a realistic and responsive experience that mimics actual movement and interaction within virtual settings.

1.2 Intended Audience

The intended audience for the VR wheelchair system includes individuals who use wheelchairs, whether due to temporary or permanent mobility impairments. This product is designed for both new wheelchair users who need to develop basic navigation skills and experienced users who wish to refine and expand their capabilities. Additionally, rehabilitation centers, physical therapy clinics, and organizations providing support for wheelchair users can utilize this system for training and educational purposes. The system is adaptable to various user needs and skill levels, making it a versatile tool for a wide range of applications in both individual and institutional contexts.

1.3 Key Requirements

- Virtual Environments: The system includes a variety of carefully designed VR environments that simulate different real-life scenarios. These environments provide users with opportunities to practice and enhance their navigation skills in a safe and controlled setting.
- User Interface: The VR system features an intuitive user interface that allows users to easily interact with the virtual environment. The interface is designed to be accessible and user-friendly, ensuring that wheelchair users can navigate the VR world with ease.
- Sensor-Based Modules: The system utilizes sensor-based modules that attach to the wheels of the wheelchair. These sensors capture the movement of the wheels and translate it into corresponding actions within the VR environment. This allows users to control their movement in the VR world by spinning the wheels in real life.
- Customization and Adaptability: The system can be customized to meet the specific needs of individual users. This includes adjusting the sensitivity of the sensors, the difficulty of the VR environments, and other settings to ensure an optimal training experience.

2 System Overview

The AccessVR system is simple and straightforward, consisting of three main layers: the Wheelchair Layer, for user input and data collection; the Data Processing Layer, for data collation and calculation; and the Headset Layer, for data interpretation and output. The system is streamlined and largely one-directional, beginning with the wheelchair and ending with the virtual environment in the headset.

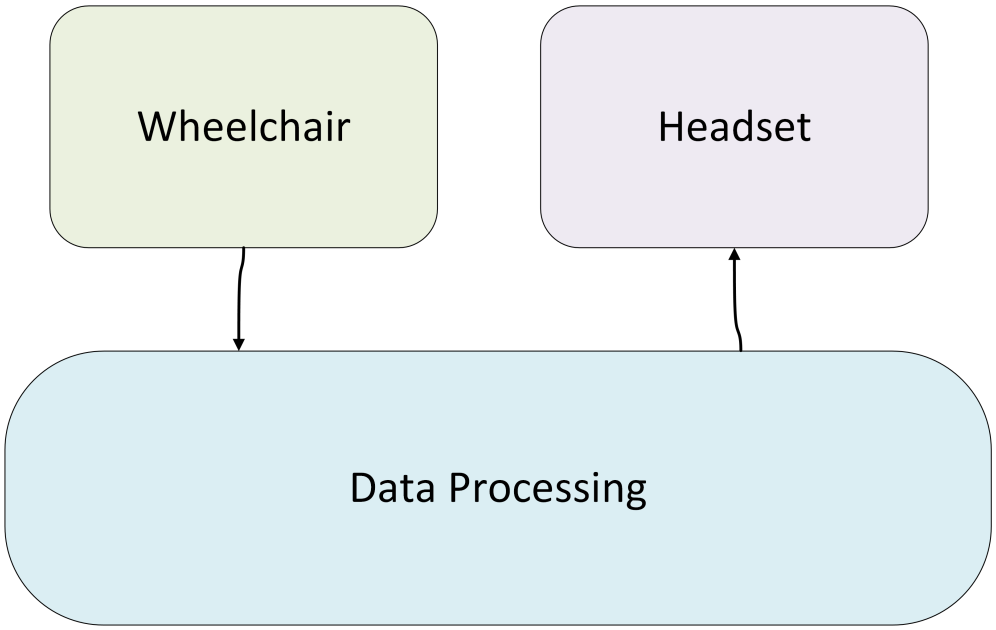


Figure 1: A simple architectural layer diagram

2.1 Headset Layer Description

The Headset layer is the layer that connects the user with the virtual environment. The player movement provides the user the ability to move around the environment. Hand tracking give the user the ability to control the UI and therefore interact with the virtual environment.

2.2 Data Processing Layer Description

The Data Processing layer is an intermediate layer connecting the wheelchair controller with the headset. The Arduino microcontroller receives all input data from the wheelchair, and sends that data to the API. The API then translates the data for use by the headset systems.

2.3 Wheelchair Layer Description

The Wheelchair layer is dedicated to the hardware aspect of this product. Which is all connected to the wheelchair so that information can be transferred to the data processing layer. Key functions:

- User controller: Movement will be dependent on the movement of the wheels and buttons.
- Data transmission: Movement will be captured from the wheel and transferred to the micro-controller and or data processing layer.

3 Subsystem Definitions & Data Flow

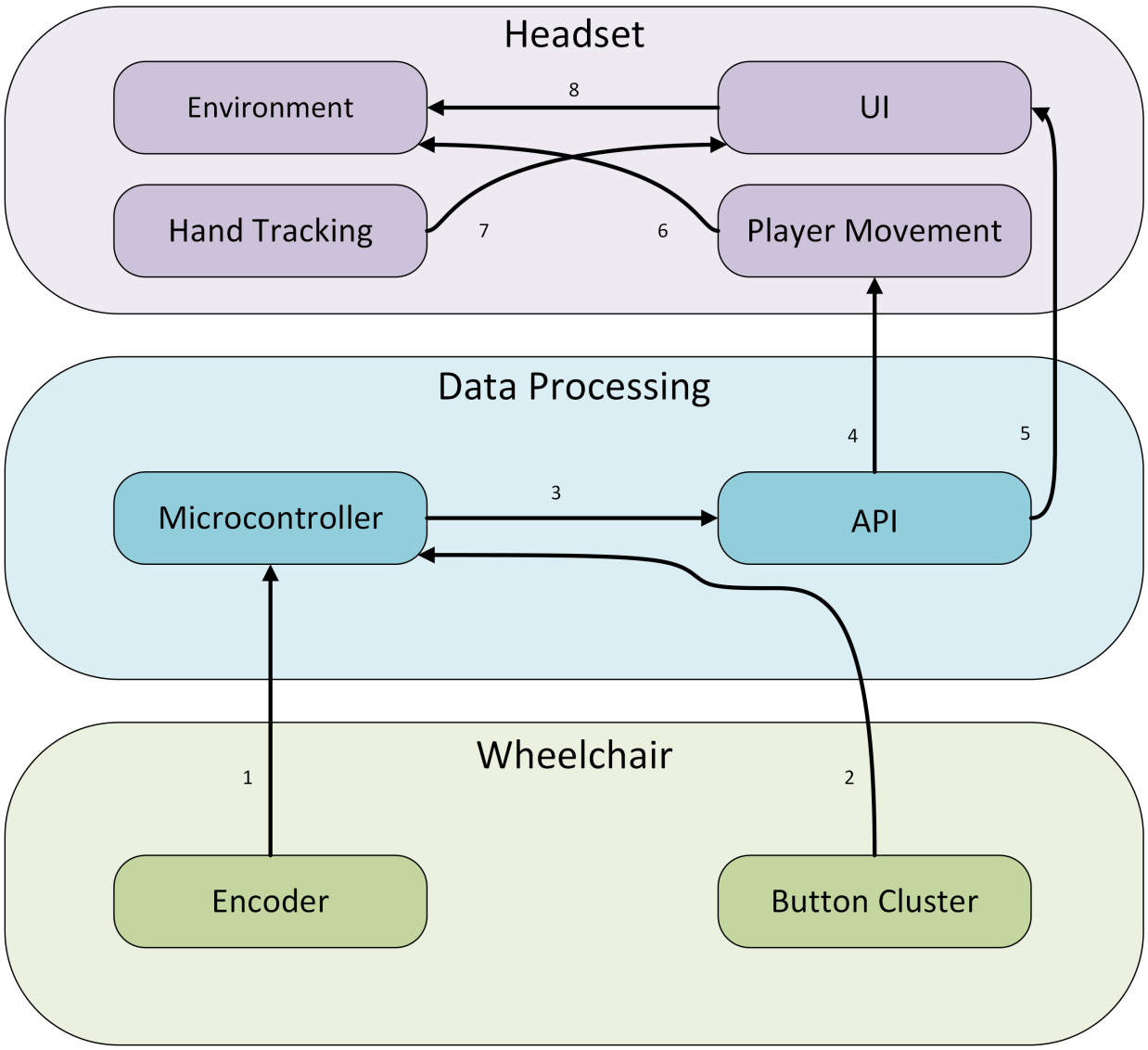


Figure 2: A simple data flow diagram

4 Headset Layer

The Headset layer is used to show the virtual environment to the user, allowing them to interact with the UI through the Meta Quests Hand Tracking Capabilities and The data processed by the Data processing layer. This data will be translated to controller input movement in Unreal Engine.

4.1 User Interface

The user interface is what the user user to interact with the virtual environment. The menu Home User interfaces will reset the player view and allow the user to change different maps.

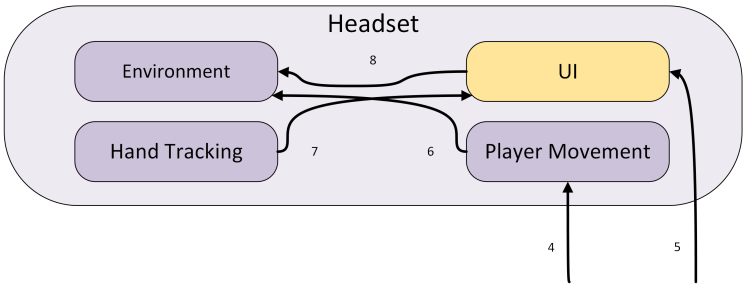


Figure 3: Example subsystem description diagram

4.1.1 Assumptions

Three assumptions are made:

- Hands: Hands will handle UI interactions to the Meta Quest 3 VR headset using their front cameras and Meta XR UE plugin.
- Menu UI: It is assumed that the player will have a menu screen.

4.1.2 Responsibilities

List of responsibilities:

- Change Settings: Menu allows user to change maps and recenter their view.
- World Interaction: UI allows user to interact with the virtual environment.

4.1.3 Subsystem Interfaces

Table 2: Subsystem interfaces

ID	Description	Inputs	Outputs
#01	Menu Screen	Left Hand Right Hand	Change Settings
#02	Game UI	Left Hand Right Hand Right wheel Left Wheel	User Locomotion

4.2 Player Movement

The player will navigate various environments using a physical wheelchair as a controller input for a wheelchair-bound avatar. The wheelchair bound avatar will be able to go up stairs, hills, and other obstacles with ease.

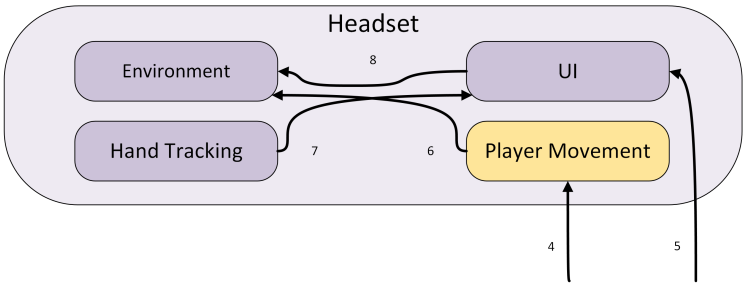


Figure 4: Example subsystem description diagram

4.2.1 Assumptions

Three assumptions are made:

- Wheels: The wheels will replace joystick locomotion in the Unreal Simulation.
- Navigation: Wheelchair avatar will be able to traverse all stairs and obstacles with ease.
- Hands: Hand tracking will allow the player to see where His hands are in respect to the wheelchair.

4.2.2 Responsibilities

List of responsibilities:

- Locomotion: Moving Around the environment and adjusting the player according to his angle to the environment to traverse the navigational assets within each environment.

4.2.3 Subsystem Interfaces

Table 3: Subsystem interfaces

ID	Description	Inputs	Outputs
#01	Locomotion for the player	Wheels Hand Tracking	Player Movement in Environment

4.3 Environment

The Virtual environments will be built using various blender made assets. Mainly Navigational such as various kinds of stairs. There will be multiple maps for the user to navigate through such as a Library, classroom, and various hills.

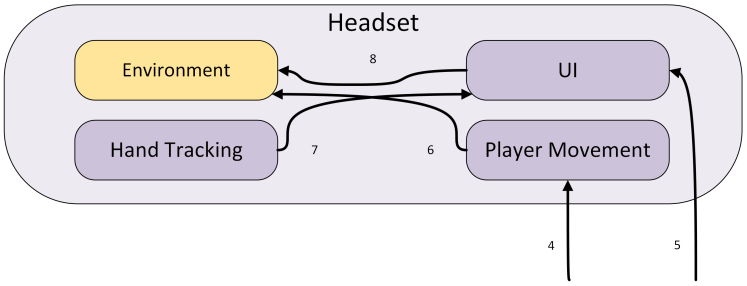


Figure 5: Example subsystem description diagram

4.3.1 Assumptions

Three assumptions are made:

- Asset Creation: Assets will be made with navigation in mind.
- Navigation: The wheelchair will be able to navigate through stairs, hills, and other obstacles.
- Base World: The library will be the main map used during demos.

4.3.2 Responsibilities

List of responsibilities:

- Maps: Showing various environments to the user.
- Navigation: Providing various navigation paths for traversal.

4.3.3 Subsystem Interfaces

Table 4: Subsystem interfaces

ID	Description	Inputs	Outputs
#01	Virtual Environments	N/A	Different maps

4.4 Hand Tracking

The Hand Tracking will be used show the players hands in the virtual environment. This will allow the user to better identify their location in respect to the world. Additionally, this is how the user will interact with their menu and the world around them.

4.4.1 Assumptions

Three assumptions are made:

- Controllers: It is assumed that no motion controllers will be used in the application.
- Wheels: It is assumed that hand tracking will simulate grabbing the virtual wheelchair's wheels when the player grabs the real ones.

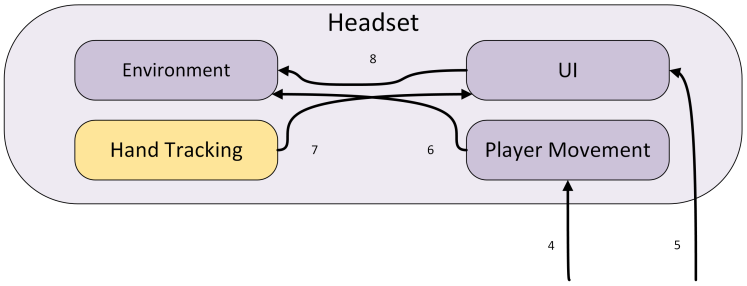


Figure 6: Example subsystem description diagram

4.4.2 Responsibilities

- Hands: Act like the users hands in the virtual environment.
- Environment Interaction: Hands will be able to press buttons or interact with the environment around it in different ways.
- Menu Interaction: Hands will be able to interact with the menu via a hand pointer or other hand tracking solution.

4.4.3 Subsystem Interfaces

Table 5: Subsystem interfaces

ID	Description	Inputs	Outputs
#01	Interacting with the environment	Left Hand Right Hand	Interacting with the environment (buttons,grabbing Items)
#02	Interact with the menu	Left Hand Right Hand	selecting menu items

5 Data Processing

The data processing layer is the bridge between the hardware components, the wheelchair, and the software side, VR headset. It collects raw data as input and transmit to the headset. This layer consists two subsections, mircoprocessor and API.

5.1 Microprocessor

The microprocessor subsystem of the VR Wheelchair System is centered around the Elegoo R3 microcontroller. This subsystem is responsible for receiving real-time data from rotary encoders attached to the wheelchair wheels. The rotary encoders measure the rotation and speed of the wheels, providing critical movement data. The Elegoo R3 processes this raw data, converting it into a format suitable for VR head translation. The processed data is then passed to the API interfacing with the Meta Quest 3 VR headset, ensuring accurate and responsive movement within the virtual environment. The subsystem includes interfaces for sensor data acquisition, data processing, and communication with the VR API, forming a crucial link in translating physical wheelchair movements into virtual actions.

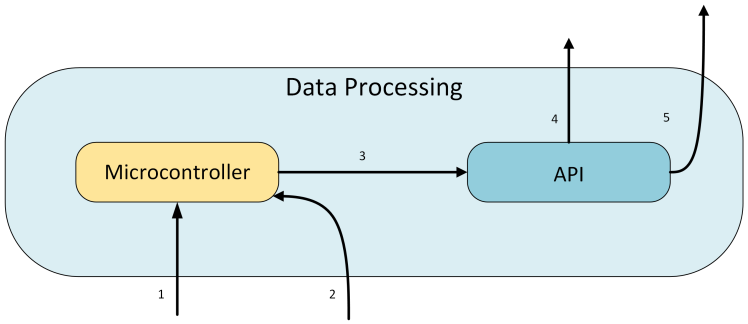


Figure 7: Example subsystem description diagram

5.1.1 Assumptions

Three assumptions are made:

- Interface Compatibility: It is assumed that the microprocessor will interface seamlessly with the sensor modules and the Meta Quest 3 VR headset.
- Processing Capacity: The microprocessor will have sufficient processing power to handle real-time data from the sensors without significant latency.
- Power Supply: The microprocessor will have a stable power supply capable of supporting continuous operation during VR sessions.

5.1.2 Responsibilities

List of responsobilities:

- Data Acquisition: Collect real-time data from the sensors attached to the wheelchair (e.g., rotary encoders).
- Data Processing: Process the raw sensor data to extract meaningful movement information, such as speed, direction.
- Data Transmission: Transmit the processed data to the Meta Quest 3 VR headset using Bluetooth or another wireless communication protocol.

5.1.3 Microprocessor Interfaces

Table 6: Subsystem interfaces

ID	Description	Inputs	Outputs
#01	Collects data from the encoder	wheel rotation	N/A
#02	Converts raw sensor data into actionable movement input	N/A	Processed movement data (speed, direction)
#03	Sends the processed movement data to the Meta Quest 3 VR headset via Bluetooth	N/A	Processed movement data

5.2 API

The API subsystem is a critical component of the data processing layer in the VR Wheelchair System. OpenXR API will be used in this scenario. This subsystem acts as an intermediary between the microprocessor (Elegoo R3) and the Meta Quest 3 VR headset. The API is responsible for receiving processed data from the microprocessor, which includes movement data derived from rotary encoders on the wheelchair wheels. It translates this data into VR head translation commands, ensuring that physical movements of the wheelchair are accurately reflected in the virtual environment. The API also handles data synchronization, error handling, and communication protocols necessary for seamless integration with the VR system.

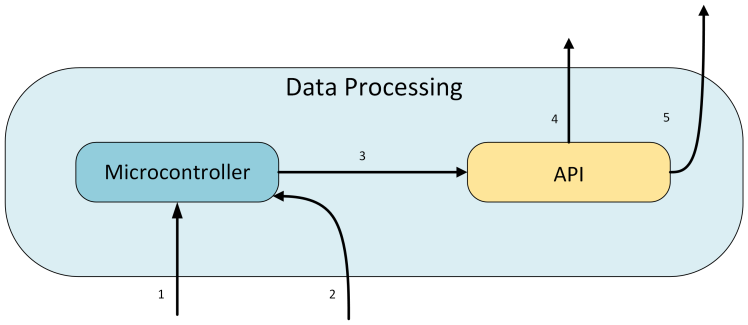


Figure 8: Example subsystem description diagram

5.2.1 Assumptions

Three assumptions are made:

- Data Integrity: Assumes that the data received from the microprocessor is accurate and timely.
- Assumes the use of a reliable wireless communication protocol (e.g., Bluetooth) between the microprocessor and the VR headset.
- Assumes that the system can handle real-time data processing and transmission without significant latency.

5.2.2 Responsibilities

List of responsobilities:

- Data Reception: Receives processed movement data from the microprocessor.
- Data Translation: Converts the movement data into VR head translation commands.
- Data Transmission: Sends the translated commands to the Meta Quest 3 VR headset.

5.2.3 API Interfaces

Table 7: Subsystem interfaces

ID	Description	Inputs	Outputs
#01	Receives data from the microprocessor indicating the wheelchair's movement.	Processed movement data (speed, direction)	N/A
#02	Sends translated movement commands to the Meta Quest 3	N/A	VR head translation commands

6 Wheelchair Layer

The Wheelchair Layer provides the primary method of user interaction with the virtual environment. The user will control navigation in the environment by turning the wheelchair wheels, and can interact with metasystems (like the UI) via use of two buttons on the wheelchair. The entire layer feeds into the microprocessor in the Data Processing layer for interpretation of the data.

6.1 Encoder

The wheelchair will have a rotary encoder attached to each wheel. These encoders will serve to track wheel movement, which can later be translated into player movement in the VR environment.

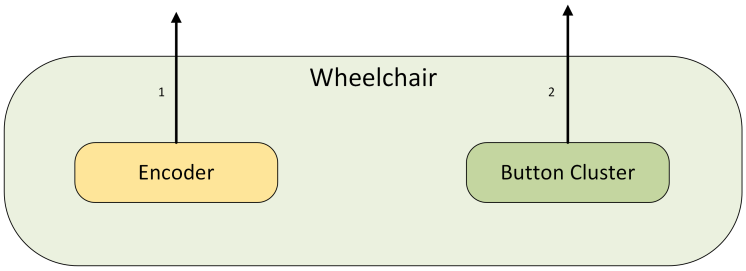


Figure 9: Example subsystem description diagram

6.1.1 Assumptions

It is assumed that the rotary encoders will be able to precisely track wheel movement and direction at all speeds, from slight movements to the maximum speed of a human propelling the wheels with their arms.

6.1.2 Responsibilities

The rotary encoders are attached to the wheelchair wheels via a system of plates and bolts which thread through the spokes of the wheels. The rotating knob of the encoder is embedded in the outermost plate and turns as the wheels turn. The weight of the encoder and wiring keep the exterior stable, allowing the knob to turn freely. As the knob turns, the encoder sends signals to the microprocessor which track the position of the encoder. When the wheels move forward, the encoder position increments. When the wheels move backward, the encoder decrements. This provides a basis for the processor to calculate movement direction and velocity.

6.1.3 Subsystem Interfaces

Table 8: Subsystem interfaces

ID	Description	Inputs	Outputs
#1	Encoder Position	User-provided (manual movement of wheels)	User & Phase B signals to processor

6.2 Button Cluster

A single button attached to each arm of the wheelchair will provide an additional method of interaction with the headset.

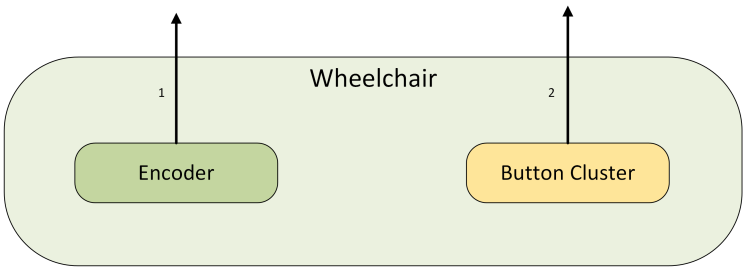


Figure 10: Example subsystem description diagram

6.2.1 Assumptions

It is assumed the button will reliably transmit data without interference to the program. Additionally, it is assumed that these buttons will be durable enough to withstand regular use and wear and tear.

6.2.2 Responsibilities

One button will be used to reset the position of the user in the environment back the starting position. The other button will bring up the user interface and menu. This is to increase accessibility and allow for the user to not need the traditionally controller. These buttons will be attached to the arm rest of the wheelchair for easy access.

6.2.3 Subsystem Interfaces

Table 9: Subsystem interfaces

ID	Description	Inputs	Outputs
#2	Buttons	User-provided (pressing of button)	Signals to processor

References